Hamiltonian Simulation in the Low-Energy Subspace

Quantum Algorithms — Spring 2026 Project Report

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Introduction: Overview and Motivations

Quantum Simulation

Simulating quantum dynamics promises breakthroughs in:

- Many-body physics
- Quantum chemistry
- Material science

Key Insight

Physical systems *typically evolve within a low-energy subspace*.

If the initial state is supported primarily in a low-energy sector, we should exploit this structure algorithmically.

The Bottleneck

Digital Hamiltonian simulation via **Suzuki–Trotter product formulas** has simulation error scaling with the **global spectral norm** $\|H\|$ of the full Hamiltonian. For large interacting systems, this norm can be **overwhelming or even unbounded**, making simulations prohibitively expensive — even if the system never visits high-energy states.

Proposed Solution

Pivot to simulating only the **physically relevant regime**: the **low-energy subspace**.

- Complexity scales with a *localized effective norm*
- Exponential suppression of leakage

Introduction: Papers Studied

Şahinoglu & Somma [2021]

If an initial state is supported in a low-energy sector, simulation complexity **drops significantly**. Probability of leaking to high-energy subspaces decays **exponentially** with the energy gap. Algorithmic complexity scales only with the localized, effective low-energy norm.

Arad, Kuwahara & Landau [2016]

Area law connection: Global and local energy distributions in quantum spin lattices are deeply connected. Truncated local Hamiltonians possess low-energy spectra exponentially close to the exact global Hamiltonian. This allows for efficiently simulating 1D and 2D spin systems.

This Project

- Synthesise the above key results
- Recreate numerics for 2×6 spin ladder
- **New**: Numerical Simulations of 8-spin chain with higher-order approximations

k-Local Hamiltonians

k-Local Hamiltonian

A Hamiltonian is k-**local** if it can be expressed as a sum of terms each acting non-trivially on at most k spins (k constant, independent of N):

$$H = \sum_{\alpha} h_\alpha, \qquad h_\alpha \text{ acts on } \leq k \text{ spins} \tag{1}$$

Each local term has **bounded strength**: $\|h_\alpha\| \leq J$, where J sets the characteristic energy scale.

Degree d

The degree d = maximum number of local terms involving any given spin. No spin participates in an unbounded number of interactions.

Consequence:

$$\text{number of terms} \leq dN \tag{2}$$

Physical Scope

Captures a broad class of physically relevant systems:

- Spin chains
- Lattice models
- Short-range interactions
- Long-range interactions

Hamiltonian Decomposition & Time Evolution

Consider a time-independent Hamiltonian expressed as a sum of L Hermitian operators:

$$H = \sum_{l=1}^{L} H_l \tag{3}$$

The corresponding time-evolution operator is:

$$U(t) = e^{-itH} = e^{-it\sum_{l=1}^{L} H_l} \tag{4}$$

Implementation Challenge

Directly implementing this unitary is difficult because exponentiating a sum of **non-commuting operators** leads to high circuit complexity.

Solution: Product Formulas

Approximate the global evolution using product formulas (Lie–Trotter or Suzuki–Trotter), which decompose the evolution into a sequence of exponentials of individual Hamiltonian terms over short time intervals. Each such exponential is typically **much easier to implement on quantum hardware**.

Nested Commutator Norm Scaling

Claim

A nested commutator of depth k satisfies $\|C_k\| = O(h^k)$.

Proof by Induction

Base case ($k = 1$):

$$\|[A, B]\| = \|AB - BA\| \leq 2\|A\|\|B\| \leq 2h^2 = O(h^2) \tag{5}$$

Inductive step: Assume $\|C_k\| = O(h^k)$. Then:

$$C_{k+1} = [A, C_k] \quad \Rightarrow \quad \|C_{k+1}\| \leq 2\|A\|\|C_k\| \leq 2h \cdot O(h^k) = O(h^{k+1}) \tag{6}$$

By induction: $\|\text{nested commutator of depth } k\| = O(h^k)$.

Product Formulas: Definition & Order

Let $W_p(s)$ denote a product formula approximation of order p for $U(s)$, $s \in \mathbb{R}$:

$$W_p(s) = e^{-is_q H_{l_q}} \cdots e^{-is_2 H_{l_2}} e^{-is_1 H_{l_1}} \tag{7}$$

where each coefficient $s_j \propto s$ and $1 \leq l_j \leq L$. The **total effective evolution time** is:

$$|s| = \sum_{j=1}^{q} |s_j|, \quad L,\, q = O(1) \tag{8}$$

p-th Order Condition

A product formula $W_p(s)$ is p-**th order** if:

$$\|U(s) - W_p(s)\| = O\big((h|s|)^{p+1}\big) \tag{9}$$

where $h = \max_l \|H_l\|$. Proof in **Appendix**

BCH Characterisation

$W_p(s)$ reproduces e^{-isH} up to all terms of order s^p. The **first non-vanishing correction** appears at order s^{p+1}, composed of nested commutators of depth $p+1$.

Higher-Order Construction

Higher-order formulas are constructed by **Suzuki's recursive method**: symmetrizing and combining lower-order approximations to cancel lower-order commutator contributions.

Trotterization Over Finite Time

To simulate evolution over a finite time t, divide the interval into r equal segments:

$$s = \frac{t}{r} \tag{10}$$

The full evolution operator is then approximated as:

$$U(t) \approx \left(W_p(s)\right)^r \tag{11}$$

Trotter Number r

r governs the trade-off between accuracy and computational cost.

- Larger $r \Rightarrow$ smaller step s
- Smaller $s \Rightarrow$ better approximation
- More expensive computation

Total Gate Count

If each exponential $e^{-is_j H_{l_j}}$ is itself decomposed into M elementary operations, total operations $\propto$:

$$qMr = O(Mr) \tag{12}$$

Overall complexity is governed by r and p.

Trotterization Over Finite Time

Limitation of Standard Bounds

Standard error bound depends on $h = \max_l \|H_l\|$, which can be **extremely large**, yielding highly pessimistic estimates. These bounds can be significantly improved when the initial state is restricted to a **low-energy subspace**.

Improved Error Bounds for Low-Energy States

Low-Energy Initial State Assumption

Suppose the initial state satisfies:

$$\Pi_{\leq \Delta} |\psi\rangle = |\psi\rangle \tag{13}$$

where $\Pi_{\leq \Lambda}$ denotes the projector onto eigenstates of H with eigenvalues $\leq \Lambda \geq 0$, and each H_l is **positive semidefinite**.

Under this assumption, the dynamics are confined to a subspace of limited energy.

Effective Operators

For a Hermitian operator X and cutoff $\Delta_0 \geq \Delta$, define the **effective operator**:

$$\bar{X} = \Pi_{\leq \Delta_0} X \Pi_{\leq \Delta_0} \tag{14}$$

Construct the effective Hamiltonian $\bar{H} = \sum_l \bar{H}_l$ and effective time-evolution $\bar{U}(s) = e^{-is\bar{H}}$.

Improved Error Bounds for Low-Energy States

Key Identity

Since the initial state lies entirely in the low-energy subspace:

$$U(t)\,|\psi\rangle = \bar{U}(t)\,|\psi\rangle \tag{15}$$

The effective norm satisfies $\bar{h} = \max_l \left\|\bar{H}_l\right\| \geq \Delta_0$, giving improved error scaling $O\big((\Delta_0 s)^{p+1}\big) \ll O\big((h|s|)^{p+1}\big)$ when $\Delta_0 \ll h$.

Decomposition into Commuting Layers

Reorganise H into a sum of groups of **mutually commuting** terms:

$$H = \sum_{l=1}^{L} H_l, \qquad H_l = \sum_{j=1}^{M} h_{l,j} \tag{16}$$

Key Property

All terms *within* a given group commute:

$$[h_{l,i},\ h_{l,j}] = 0 \quad \forall\, i, j \tag{17}$$

while terms in *different* groups may not commute.

Decomposition into Commuting Layers

Size Constraints

- $L = O(1)$ for constant d and k
- $LM \leq dN$

Exact Decomposition Within Groups

Since $[h_{l,i}, h_{l,j}] = 0$:

$$e^{-iH_l t} = \prod_{j=1}^{M} e^{-ih_{l,j} t} \tag{18}$$

This is **exact** (no Trotter error within a group).

Implications

Decomposition enables Trotter–Suzuki approximations where errors arise **only from non-commutativity between different groups**. Reduces many-body evolution to sequences of few-body evolutions efficiently implementable on quantum hardware.

High-Energy Leakage: Setup & the Leakage Bound

Consider $H = \sum_{l=1}^{L} H_l$ with each H_l local and positive semidefinite.

- $\Pi_{\leq\Delta}$: projector onto eigenstates of H with energy $\leq \Delta$ (max energy of initial state)
- $\Pi_{>\Lambda_0}$: projector onto eigenstates with energy $> \Lambda_0$, where $\Lambda_0 \geq \Delta$
- Initial state: $\Pi_{\leq\Delta} |\psi\rangle = |\psi\rangle$

Leakage Bound (Arad, Kuwahara & Landau)

For local Hamiltonians, the **leakage** to high-energy states under a single local-term evolution is bounded by:

$$\left\| \Pi_{>\Lambda_0} \, e^{-isH_l} \, \Pi_{\leq\Delta} \right\| \;\leq\; e^{-\alpha_1(\Lambda_0-\Delta)/J} \left(e^{\alpha_2 J|s|N} - 1 \right) \tag{19}$$

for strict positive constants $\alpha_1, \alpha_2 > 0$. Here $J = \max_X \|h_X\|$ is the energy scale.

Energy Gap Suppression

Factor $e^{-\alpha_1(\Lambda_0-\Delta)/J}$: transitions to higher-energy states are **exponentially suppressed** by the size of the energy gap.

Local Spreading

Factor $e^{\alpha_2 J|s|N}$: repeated application of local operators increases the spatial support of the state, allowing energy to spread across the lattice.

Main Theorem: Setup & Statement

Setup & Notation

- $H = \sum_{l=1}^{L} H_l$: local, positive semidefinite terms
- $\Pi_{\leq \Delta} |\psi\rangle = |\psi\rangle$: initial state in low-energy sector
- Δ: maximum intrinsic energy of initial state
- $\Delta_0 \geq \Delta$: effective low-energy spectral cutoff
- $U(s) = e^{-isH}$: exact time-evolution; $W_p(s)$: p-th order product formula

Main Theorem: Setup & Statement

Theorem: Low-Energy Product Formula Error (Şahinoglu & Somma)

For a local Hamiltonian and a p-th order product formula, the approximation error satisfies:

$$\left\| \left(U(s) - W_p(s)\right)\Pi_{\leq\Delta} \right\| = O\left((\Delta_0|s|)^{p+1}\right) \tag{20}$$

where the **effective energy scale** is:

$$\boxed{\Delta_0 = \Delta + \beta_1 J \log\frac{\beta_2}{J|s|} + \beta_3 J^2 N|s|} \tag{21}$$

and $\beta_1, \beta_2, \beta_3 > 0$ depend on system geometry.

Main Theorem: Interpretation

Three Physical Contributions to Δ_0

1. Δ — **Intrinsic energy** of the initial state.
2. $\beta_1 J \log(\beta_2/(J|s|))$ — **Mathematical correction** from balancing leakage errors; arises from optimal choice of cutoff.
3. $\beta_3 J^2 N|s|$ — **Accumulated energy growth** caused by local interactions spreading across N spins over time s.

Main Theorem: Interpretation

Key Improvement

When $\Delta \ll h$, we have $\Delta_0 \ll h$, and so:

$$(\Delta_0|s|)^{p+1} \ll (h|s|)^{p+1} \tag{22}$$

The error bound represents a **massive improvement** over standard worst-case bounds.

Key Idea Behind Proof

Two ingredients:

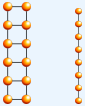
1. **Effective Hamiltonian**: restrict to low-energy subspace, reducing operator norm from h to Δ_0.
2. **Leakage bound**: replacing effective evolution with true evolution introduces only exponentially small error.

Together: product formula behaves as if global norm were Δ_0.

Numerical Setup: Spin Systems & Hamiltonian

Two Spin Systems Studied

1. 2×6 **Heisenberg ladder** (12 spins, Hilbert space $2^{12} = 4096$)
   - Horizontal nearest-neighbour couplings along each row
   - Vertical couplings connecting the two rows
   - Sparse matrix representations



2. **1D Heisenberg chain** (8 spins, Hilbert space $2^8 = 256$)
   - Nearest-neighbour interactions
   - Dense matrix representations

Heisenberg Hamiltonian

$$H = -\sum_{\langle i,j \rangle} (X_i X_j + Y_i Y_j + Z_i Z_j) \tag{23}$$

Properties:

- Isotropic interaction (same x, y, z)
- Ferromagnetic coupling (energy minimised when spins align)
- $H_{ij} \propto -\vec{S}_i \cdot \vec{S}_j$

Numerical Setup: Spin Systems & Hamiltonian

Hamiltonian Decomposition (Both Systems)

$$H = H_1 + H_2 \tag{24}$$

For the 1D case, Interaction terms alternated so that each H_l contains **local interactions**: even bonds $\rightarrow H_1$, odd bonds $\rightarrow H_2$.
For the 2D case, Interaction terms are arranged such that all the horizontal bonds and alternate vertical bonds are in H_1, and the other set of alternate vertical bonds are in H_2.

Low-Energy Subspace Construction & Error Measures

Subspace Construction

The Hamiltonian is diagonalised to obtain eigenvalues and eigenvectors. Lowest-energy eigenstates define a sequence of subspaces:

$$\mathcal{H}_n = \text{span}\{|E_1\rangle, \ldots, |E_n\rangle\} \tag{25}$$

Errors evaluated for $n = 1$, 50, 100, 150, 200 — systematic study of how accuracy depends on energy scale.

Low-Energy Subspace Construction & Error Measures

Product Formulas Tested

1st Order (Lie–Trotter):

$$W_1(s) = e^{-isH_1} e^{-isH_2}$$

2nd Order (Strang Splitting):

$$W_2(s) = e^{-isH_1/2} e^{-isH_2} e^{-isH_1/2}$$

4th Order (1D chain only):

$$W_4(s) = W_2(p_1 s)^2 \, W_2(p_0 s) \, W_2(p_1 s)^2 \tag{26}$$

with p_0, p_1 chosen for 4th-order accuracy.

Error Measures

Worst-Case Error:

$$\|U(s) - W(s)\| \tag{27}$$

Operator norm over full Hilbert space.

Low-Energy Error:

$$\|(U(s) - W(s))\, P_n\| \tag{28}$$

where P_n projects onto $\mathcal{H}_n$.

Results: 2×6 Ladder System

Observed Behaviour

- Worst-case error **grows rapidly** with time step s
- Errors restricted to low-energy subspaces are **significantly smaller**
- 2nd-order formula reduces error **substantially** compared to 1st-order
- The dashed (worst-case) curves lie well above solid (low-energy) curves at all time steps

System Characteristics

- **Higher connectivity** than 1D chain $\Rightarrow$ larger operator norms and generally higher errors
- **12 spins**: increased complexity and error accumulation
- Study includes up to **2nd-order** approximation

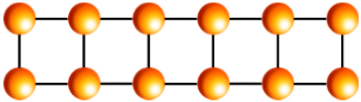


Figure: 2x6 Ladder Chain

Key Observation

Low-energy errors grow **much more slowly** than worst-case errors, confirming worst-case bounds overestimate practical simulation errors.

Results 2×6 Ladder System

Plots

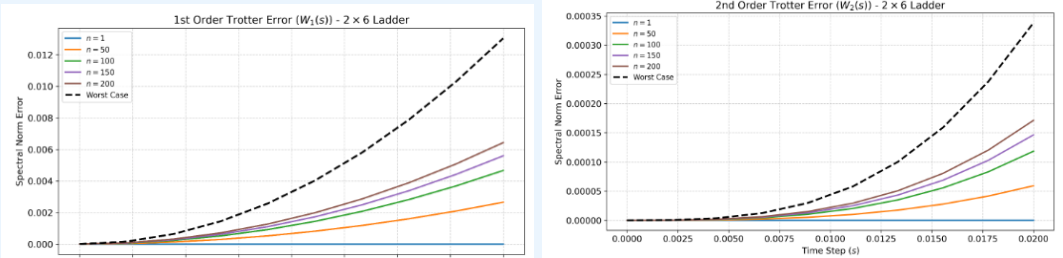


Figure: Plots of the Code for the 2×6 ladder system

Results: 1D Chain (8 Spins)

Observed Behaviour

- **Similar separation** between worst-case and low-energy errors as ladder
- Higher-order formulas (2nd and 4th order) **significantly improve** accuracy
- 4th-order method achieves **very small errors** even for larger time steps
- Errors grow much more slowly in the low-energy subspace

Key Differences vs. Ladder

- **Lower connectivity** $\Rightarrow$ smaller operator norms
- **Smaller system** (8 vs. 12 spins) $\Rightarrow$ less complexity
- **Higher order available**: 4th-order clearly demonstrates improved scaling

Figure: 1x8 Spin System

Conclusion of Numerics

Trotter errors significantly reduced in low-energy subspaces.
Effect consistent across geometries, more pronounced with higher-order product formulas.
Confirms: low-energy dynamics can be simulated more efficiently than worst-case bounds suggest.

Results 1×8 Spin System

Plots

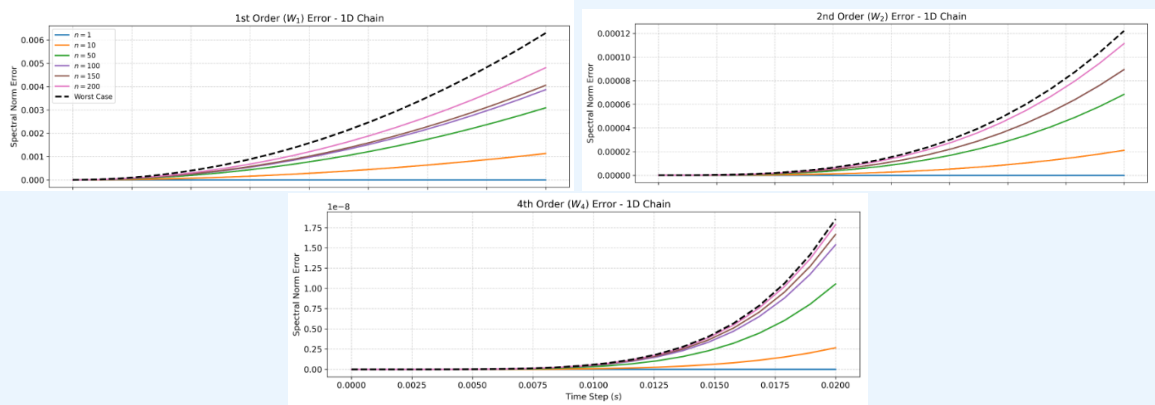


Figure: Plots of the Code for the 1×8 spin system

Positivity of Local Terms: Physical Relevance

The Positivity Requirement

The error bounds require each local term to satisfy:

$$H_l \geq 0 \qquad \text{(positive semidefinite)} \tag{29}$$

Under this condition, for any low-energy state $|\psi\rangle$ (energy $\leq \Delta_0$):

$$\langle\psi| H_l |\psi\rangle \leq \langle\psi| H |\psi\rangle \leq \Delta_0 \quad \Rightarrow \quad \left\|\bar{H}_l\right\| \leq \Delta_0 \tag{30}$$

This bound is **crucial** for obtaining improved error estimates.

Positivity of Local Terms: Physical Relevance

Shifting to Enforce Positivity

Can always shift: $H_l \to H_l + a_l I$ to make the smallest eigenvalue zero. **Danger**: if shift scales with N, it increases Δ (and hence Δ_0) proportionally, **destroying the advantage** of the low-energy approach.

Natural Positivity

Many physical Hamiltonians satisfy $H_l \geq 0$ **without** artificial shifts!

- Heisenberg models
- Anisotropic Heisenberg models
- Frustration-free Hamiltonians

Effective Low-Energy Description & Area Laws

- In many-body physics and quantum information theory, an *area law* refers to the scaling behaviour of information or entropy associated with a spatial region.
- In its general form, an area law states that the information content of a region (in particular, its entanglement entropy or total correlations) scales with the *area of the boundary* of the region rather than with its volume.

The area law therefore states that

$$S(A) \propto |\partial A|$$

for systems with finite correlation length.

Why Area Laws Occur

- The intuitive reason behind this behaviour is that in physical systems with *short-ranged correlations*—that is, systems with finite correlation length ξ—degrees of freedom deep inside the region A are effectively uncorrelated with those outside.
- Only the degrees of freedom lying within a strip of width $\sim \xi$ from the boundary ∂A can be strongly correlated with the exterior.
- Consequently, the amount of information shared between A and its complement B is proportional to the number of degrees of freedom on the boundary, i.e. proportional to $|\partial A|$.

Effective Low-Energy Description & Area Laws

Effective Low-Energy Description

With projector $\Pi_{\leq\Delta}$ and $\Delta_0 \geq \Delta$, define:

$$\bar{H} = \Pi_{\leq\Delta_0} H \Pi_{\leq\Delta_0} \tag{31}$$

Dynamics generated by $\bar{H}$ agree with H on low-energy states up to small errors. A truncated Hamiltonian $\tilde{H}$ (high-energy contributions in a local region cut off) has low-energy spectrum **exponentially close** to the original — demonstrating stability under truncations.

Unified Perspective: Area Laws & Low-Energy Simulation

Ground state Approximation

The truncation procedure ensures correlations across a boundary are controlled by the strength of interactions *near* that boundary. This enables ground states to be approximated by states whose entanglement entropy obeys an area law:

$$S(\rho_A) = O(|\partial A|) \tag{32}$$

Unified Structural Feature

Both **low-energy Hamiltonian simulation** and **area-law proofs** rely on the same structural feature of local quantum systems:

An effective separation between low- and high-energy sectors, enforced by locality.

- **Dynamical**: suppressed transitions to high-energy states during time evolution
- **Static**: concentration of energy distributions and stability under local truncations

Unified Perspective: Area Laws & Low-Energy Simulation

Deeper Principle

Local Hamiltonians do not explore Hilbert space uniformly- Area laws for ground states arise from the combination of spatial locality of interactions and the presence of an energy gap, which together enforce exponential decay of correlations and constrain entanglement to boundary regions.

Conclusion

Summary of Key Results

1. Product formula error scales with **effective low-energy norm** Δ_0 — not the full spectral norm h. Formally:
$$\|(U(s) - W_p(s))\Pi_{\leq\Delta}\| = O\big((\Delta_0|s|)^{p+1}\big)$$
2. High-energy leakage from local evolution is **exponentially suppressed** by the energy gap $\Lambda_0 - \Delta$.
3. Positivity of local terms (e.g., Heisenberg, frustration-free) is a **natural and key physical property** enabling these bounds.
4. Area laws and low-energy simulation share the same structural origin: **locality in energy space**.

Numerical Confirmation

Both the 2×6 ladder and 8-spin 1D chain confirm that low-energy Trotter errors grow **far more slowly** than worst-case bounds. Higher-order formulas amplify this advantage.

Conclusion

Broader Significance

By anchoring simulation in the low-energy sector, product formulas become remarkably more efficient — the key step towards practical quantum simulation of physically relevant many-body systems.

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Appendix A: Error Scaling of Product Formulas

For $H = A + B$ (two Hermitian operators), Taylor-expand each exponential to order s^2:

$$W_1(s) = e^{-isA}e^{-isB} = I - is(A+B) - \frac{s^2}{2}(A^2 + B^2) - s^2 AB + O(s^3) \tag{33}$$

$$U(s) = e^{-is(A+B)} = I - is(A+B) - \frac{s^2}{2}(A^2 + B^2 + AB + BA) + O(s^3) \tag{34}$$

Result

$$W_1(s) - U(s) = -\frac{s^2}{2}[A, B] + O(s^3) \quad \Rightarrow \quad \|U(s) - W_1(s)\| = O(h^2 s^2) \tag{35}$$

General p-th order: all BCH terms up to s^p cancel, leading correction at s^{p+1}:

$$\|U(s) - W_p(s)\| = O(h^{p+1}|s|^{p+1}) \tag{36}$$

Appendix C: Full Leakage Bound Derivation

Taylor-expand and apply projectors:

$$\Pi_{>\Lambda_0}\, e^{-isH_l}\, \Pi_{\leq\Delta} = \sum_{n=1}^{\infty} \frac{(-is)^n}{n!} \Pi_{>\Lambda_0} (H_l)^n \Pi_{\leq\Delta} \tag{37}$$

Locality Inequality (Arad et al.)

For any operator A:

$$\|\Pi_{>\Lambda_0} A\, \Pi_{\leq\Delta}\| \leq \|A\|\, e^{-\lambda(\Lambda_0 - \Delta - 2R)} \tag{38}$$

where $\lambda = O(1/J)$ and R is the total interaction strength of terms non-commuting with A.

Applying with $A = (H_l)^n$, bounding $\|(H_l)^n\| \leq (JM)^n$ and $R \leq Jdkn$:

$$\|\Pi_{>\Lambda_0} (H_l)^n \Pi_{\leq\Delta}\| \leq (CJ)^n\, e^{-\lambda(\Lambda_0 - \Delta)} \tag{39}$$

Summing the Taylor series $\sum_{n=1}^{\infty} x^n/n! = e^x - 1$:

$$\left\|\Pi_{>\Lambda_0}\, e^{-isH_l}\, \Pi_{\leq\Delta}\right\| \leq e^{-\lambda(\Lambda_0 - \Delta)} \left(e^{CJ|s|} - 1\right) \tag{40}$$

Role of the Leakage Bound in Hamiltonian Simulation

Physical Interpretation

Although local evolution can temporarily generate higher-energy components, this effect is **strongly suppressed** when:

- The energy gap $\Lambda_0 - \Delta$ is large, **and**
- The local evolution time step $|s|$ is kept small.

Consequence for Algorithms

The leakage bound **theoretically guarantees** that simulated dynamics remain mostly localised within the low-energy subspace. Errors from rare transitions to higher-energy states are **strictly bounded and controlled**.

This allows analytically replacing the full Hamiltonian with an **effective low-energy Hamiltonian** without introducing large, unbounded simulation errors.

This forms the **backbone** for the improved error scaling in the Main Result.

Heisenberg Interaction & Frustration-Free Hamiltonians

Heisenberg Two-Spin Interaction

The Heisenberg interaction (up to additive constant):

$$h_{ij} = I - X_i X_j - Y_i Y_j - Z_i Z_j = I - \vec{\sigma}_i \cdot \vec{\sigma}_j \tag{41}$$

The operator $\vec{\sigma}_i \cdot \vec{\sigma}_j$ has eigenvalues $+1$ (triplet) and -3 (singlet), so h_{ij} has eigenvalues **0 and 4** — thus **positive semidefinite**. Each H_l is a sum of such commuting positive semidefinite operators.

Frustration-Free Hamiltonians

A Hamiltonian $H = \sum_l H_l$ is **frustration-free** if its ground state $|\psi_0\rangle$ simultaneously minimises each local term:

$$H_l |\psi_0\rangle = 0 \quad \forall\, l \tag{42}$$

Each H_l is positive semidefinite and the **ground state energy is exactly zero**.

Energy Localization in Local Hamiltonians

For a quantum spin system with local Hamiltonian $H = \sum_X h_X$, local operators **cannot efficiently generate arbitrarily large energy excitations**. If a state is supported on a low-energy subspace, its weight in high-energy sectors is **exponentially suppressed**.

Dynamical Viewpoint

In Hamiltonian simulation restricted to low-energy states, evolution under local terms produces only **exponentially small leakage** into high-energy subspaces.
$\Rightarrow$ Dynamics can be effectively described within a **truncated low-energy sector**.

Static Viewpoint

A state with bounded global energy remains concentrated within a bounded energy window when projected onto a local Hamiltonian.
Any contribution outside this window decays **exponentially** with the energy difference.

Common Principle

Local interactions constrain the redistribution of energy, leading to an **approximate decoupling between low- and high-energy sectors**.

Proof Sketch of the Main Result

We aim to bound $\|(U(s) - W_p(s))\Pi_{\leq\Delta}\|$.

Step 1 — Triangle inequality decomposition:

$$\|(U(s) - W_p(s))\Pi_{\leq\Delta}\| \leq \underbrace{\left\|(U(s) - \bar{W}_p(s))\Pi_{\leq\Delta}\right\|}_{\text{(I) effective formula error}} + \underbrace{\left\|(\bar{W}_p(s) - W_p(s))\Pi_{\leq\Delta}\right\|}_{\text{(II) leakage error}} \tag{43}$$

Step 2 — Bound (I): Since $U(s)\Pi_{\leq\Delta} = \bar{U}(s)\Pi_{\leq\Delta}$ and $\left\|\bar{H}_l\right\| \leq \Delta_0$, standard product formula bounds give:

$$\text{(I)} = O\big((\Delta_0|s|)^{p+1}\big) \tag{44}$$

Step 3 — Bound (II): Using the leakage bound with intermediate projectors, each transition outside the energy window is exponentially suppressed:

$$\text{(II)} = O(\delta), \quad \text{made small by choosing } \Delta_0 \text{ optimally} \tag{45}$$

Step 4 — Optimal Δ_0: Balancing (I) and (II) (product formula error grows with Δ_0; leakage error shrinks exponentially with $\Delta_0 - \Delta$) yields:

$$\Delta_0 = \Delta + O\left(J \log \frac{1}{J|s|} + J^2 N|s|\right) \tag{46}$$

Effective Low-Energy Description & Area Laws

Implications for Entanglement Area Laws

Area laws state that for ground states of local Hamiltonians, entanglement entropy of a region scales with its **boundary size** (not volume). The connection:

1. **Energy confinement**: Exponentially small weight in high-energy sectors.
2. **Local stability**: Truncating high-energy contributions changes the state by only exponentially small corrections.
3. **Bounded effective interactions**: Truncated Hamiltonian norm does not scale extensively with N.
4. **Restricted entanglement generation**: Local interactions + limited energy fluctuations $\Rightarrow$ long-range entanglement cannot be generated efficiently.